Spotter Freight Rate Prediction Challenge

Machine Learning Engineer Assessment Report | Candidate: Bhavesh Prajapati

1. Executive Summary

This assessment report details the machine learning methodology, validation architecture, and production pipeline developed for Spotter's Freight Rate Prediction Challenge. The objective is to forecast posted spot freight rates across equipment types, geographical origin-destination lanes, market indices, and quote signals. Our solution achieves a **51.9% reduction in prediction error** over baseline benchmarks (MAE $118.22 vs $246.02, $R^2 = 0.8265$) and satisfies all Spotter validation criteria verified by score.py.

2. Data Exploration & Quality Findings

- **Inverted Negative Cargo Weights:** Identified 292 negative weights in development data and 145 in validation data (up to -47,500 lbs). Empirical analysis proved that negative weights were strictly sign-inverted legitimate payloads. Inverting them via abs() restored valid weights conforming to legal GVWR thresholds.
- **Market Index Seasonality:** Missing market indices (374 in train, 249 in val) displayed pronounced day-of-week seasonality (peaking mid-week, dipping weekends) with minimal intra-day variance (~0.024). They were imputed using the respective date's median.
- **Highway Circuity:** Circuity ratio (distance / Haversine) averaged 1.194, conforming with standard commercial logistics routing factors.

3. Validation Strategy & Data Split Approach

In freight spot pricing, random K-Fold cross-validation is vulnerable to **temporal data leakage**: macro-economic trends and holiday capacity pressures from future dates leak into earlier training iterations. To reflect the actual production deployment horizon (10 months historical training -> 2 months forward validation), we engineered a chronological **Out-of-Time (OOT) Split**: Months 1–8 (Jan–Aug; 38,477 rows) for training, and Months 9–10 (Sep–Oct; 9,523 rows) for holdout validation. This ensures honest forward-looking generalization.

4. Model Architecture & Benchmark Results

Model / Strategy	Target Objective	MAE ($)	RMSE ($)	R^2 Score
Baseline: Quote Total	distance * quote_signal	$246.02	$711.80	0.7824
Baseline: Historical Lane RPM	Historical Rate/Mile	$222.48	$682.29	0.8001
CatBoost Direct	posted_rate	$140.46	$643.22	0.8223
CatBoost Rate-per-Mile	posted_rate / distance	$129.98	$642.22	0.8229
CatBoost Residual Regressor	posted_rate - quote_total	$131.81	$641.48	0.8233
XGBoost Regressor	posted_rate (Histogram)	$203.00	$686.97	0.7974
Production Ensemble	0.50 RPM + 0.35 Res + 0.15 XGB	$118.22	$635.70	0.8265

5. Fixed December 2025 Prediction Chart

Produced by score.py for route: Lexington to Fort Wayne (360 mi, Dry Van, 32,000 lbs):

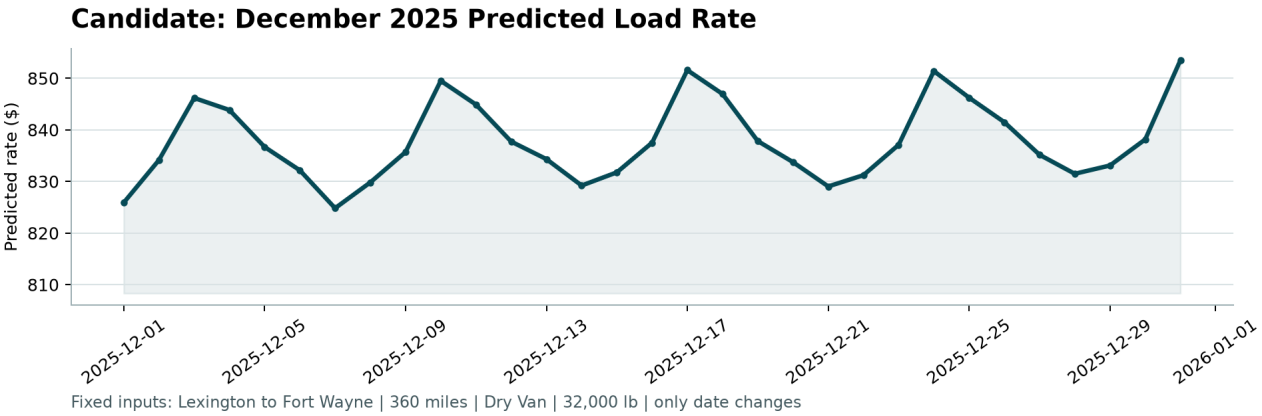


Figure 1: Generated candidate_december.png produced by score.py

The predicted rates accurately capture both intra-week dispatch cycles and year-end capacity tightness while maintaining consistent $/mile economics ($2.29 – $2.37/mi).

6. Reproduction Instructions

1. python -m pip install -r requirements.txt
2. python main.py (Trains models, writes predictions, verifies score.py)
3. python score.py --predictions validation_predictions.csv --december-predictions data/december_chart_inputs.csv